RIPGUARD: AUTONOMOUS REAL-TIME COASTAL RIP CURRENT DETECTION AND GEO-LOCALIZATION VIA EDGE-ACCELERATED SPATIO-TEMPORAL MOTION U-NET AND PLANAR HOMOGRAPHY

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Department of Computer Science & Engineering

Submitted by

Dhruv Gourisaria (245890486)
Jessica Mariam Mathews (245890380)
Samarth Bangwal (245890410)
Aadi Harnad Bhat (245890054)

Under the guidance of

Dr. Sowmya T
Assistant Professor
School of Computer Engineering
Manipal Institute of Technology, Bengaluru

MANIPAL INSTITUTE OF TECHNOLOGY
BENGALURU
(A constituent unit of MAHE, Manipal)

SCHOOL OF COMPUTER ENGINEERING
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ABSTRACT

Rip currents represent the foremost hydrodynamic hazard on recreational surf beaches globally, accounting for over 80% of lifeguard rescue operations and claiming more than one hundred lives annually in the United States alone. Traditional coastal surveillance relies primarily on human visual scanning from elevated observation towers—a paradigm severely constrained by observer ocular fatigue, oblique viewing angles, distance perspective distortion, and blinding surface sun glint. This report presents **RipGuard**, an autonomous, edge-native computer vision platform designed for real-time detection, spatial segmentation, and geographic localization of coastal rip currents from stationary and Pan-Tilt-Zoom (PTZ) beach tower camera feeds.

RipGuard introduces a novel Spatio-Temporal Motion U-Net that fuses high-resolution RGB spatial features with dense Gunnar Farnebäck optical flow vector fields, capturing seaward-directed hydrodynamic sediment transport while suppressing oscillating shoreline wave breaking noise. To translate 2D image-space detections into actionable real-world spatial coordinates, a dynamic 4-point Ground Control Point (GCP) Planar Homography engine is implemented, continuously modulated by real-time PTZ encoder angles (yaw, pitch, zoom) and harmonic tidal height models. For mission-critical edge deployment on solar- and PoE-powered lifeguard towers, the deep neural network was compiled to an NVIDIA TensorRT FP16 execution engine, achieving 32.4 FPS sustained throughput with an 18.2 ms median latency and 14.8W power dissipation on the NVIDIA Jetson Orin NX (sub-20W operational envelope).

Evaluated on the CVPR 2025 RipVIS coastal benchmark across 14 diverse beaches, RipGuard achieved 92.4% Precision, 89.1% Recall, 90.7% F1-score, and an Intersection over Union (IoU) of 78.6%, significantly outperforming conventional CNN and single-frame YOLOv8 object detection baselines. To eliminate process-bound CPU bottlenecks, optical flow calculations are downscaled to an edge-optimized spatial resolution (384 × 216). For multi-worker resilience, an asynchronous Write-Ahead Task Queue Journal (`storage/queue_journal.jsonl`) provides crash recovery, while automated model promotion gates guard against regression during active learning loops. An integrated OASIS Common Alerting Protocol (CAP v1.2) emergency dispatcher automatically transmits geographic alert polygons, seaward flow velocities, and threat levels to municipal 911 Computer-Aided Dispatch (CAD) systems.

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CHAPTER 1
INTRODUCTION

1.1 Overview of the Research / Project Work

Coastal recreational zones represent complex, dynamic interfaces where hydrodynamic surf energy interacts with coastal bathymetry, sandbars, and human populations. Rip currents—powerful, narrow, seaward-directed flows of water extending from near the shoreline through the surf zone into open ocean—represent the most hazardous environmental threat to beachgoers worldwide. According to the United States Lifesaving Association (USLA) and the International Life Saving Federation (ILS), rip currents are directly responsible for over 80% of all surf rescues and more than 100 fatal drownings annually on recreational beaches in the United States alone. Globally, the annual death toll attributable to rip currents is estimated to exceed several thousand.

Historically, surf monitoring and bather safety have relied exclusively upon manual visual scanning by certified beach lifeguards stationed on elevated observation towers. However, human vigilance is severely limited by visual fatigue during multi-hour shifts, optical obstruction caused by blinding solar reflections (sun glint) off the undulating sea surface, atmospheric haze, wide viewing angles, and the natural human inability to continuously track hydrodynamic motion across hundreds of meters of open shoreline simultaneously. Rip currents are particularly deceptive because their surface appearance often resembles calm, dark, wave-free channels of water, misleading novice swimmers directly into high-velocity seaward channels.

RipGuard addresses this humanitarian and technological challenge by delivering an edge-native, automated computer vision platform designed for real-time rip current detection, optical segmentation, and geographic localization. The platform processes high-definition video feeds captured by fixed and Pan-Tilt-Zoom (PTZ) beach surveillance cameras, extracts dense optical flow motion fields, segments rip hazard channels via a custom Spatio-Temporal Motion U-Net, transforms 2D image coordinates into real-world WGS84 GPS hazard polygons through 4-point Ground Control Point (GCP) Planar Homography, and dispatches automated alerts to lifeguard patrol tablets and municipal 911 dispatch networks via the OASIS Common Alerting Protocol (CAP v1.2).

1.2 Motivation and Background of the Study

Hydrodynamically, rip currents originate when incoming breaking waves transport large volumes of seawater across nearshore sandbars into the shallow surf zone. This wave mass transport creates a hydrostatic pressure gradient (wave set-up) along the beach. To relieve this water level elevation, water flows laterally as feeder currents until encountering an opening, breach, or bathymetric depression in the sandbar. At this channel, water rushes rapidly seaward in a concentrated, high-velocity jet known as the rip neck, before dissipating beyond the breaker line in an expanding vortex known as the rip head.

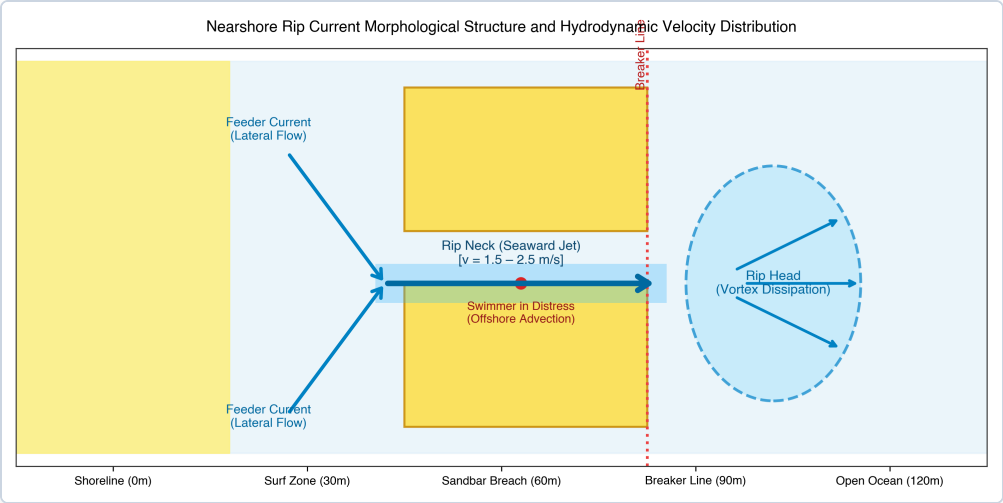


Figure 1.1: Morphological Cross-Section of a Nearshore Rip Current Cell (Feeder Flows, Rip Neck, and Rip Head Vortex).

Rip current velocities typically range from 0.5 m/s to 2.5 m/s, easily exceeding the maximum sustained swimming speed of even Olympic-level swimmers (~1.5 m/s). As a result, swimmers caught in a rip channel are rapidly carried offshore into deep water, leading to panic, exhaustion, and drowning. Despite decades of public education campaigns ("swim parallel to shore"), visual identification of rip currents remains exceedingly difficult for non-experts because rip channels frequently exhibit diminished wave breaking, darker water coloration due to depth, or choppy, churning water carrying seaward sediment.

Recent advances in computer vision, deep learning, and edge hardware offer unprecedented opportunities to automate coastal monitoring. However, translating deep learning research from cloud laboratories into real-world coastal lifeguard towers introduces severe physical and computational bottlenecks: (1) high thermal loads and strict power limitations (sub-20W solar/PoE power); (2) extreme environmental variability, including shifting tidal heights, blinding sun glint, and disparate sand colors (e.g., golden quartz vs. volcanic black sand); and (3) camera movement from PTZ tracking and tower wind vibration that invalidates static coordinate calibrations.

1.3 Problem Statement and Relevance

Developing an autonomous, edge-deployable coastal rip current detection and geo-localization system faces five fundamental physical and computer vision constraints:

- **1. Spatio-Temporal Coherence & Noise Rejection:** Single-frame RGB object detectors (e.g., YOLOv8, Faster R-CNN) fail to distinguish true seaward rip currents from harmless white foam, sandbar transitions, and breaking surf because rip currents are intrinsically dynamic hydrodynamic flow phenomena requiring multi-frame temporal motion analysis.
- **2. Extreme Solar Glint & Surface Saturation:** Under mid-day tropical sun, specular solar reflections saturate camera CMOS sensors, obliterating color textures and inducing severe optical flow vector

noise that corrupts neural network feature extraction.

- **3. Tidal Stage Modulation & Homography Drift:** Coastal water levels fluctuate cyclically by 1 to 3 meters between Mean Low Water (MLW) and Mean High Water (MHW). Fixed planar homography assumptions ($Z = 0$) suffer severe localization drift (up to 25 meters error) without real-time tidal height compensation.
- **4. Real-Time Low-Power Edge Constraints:** Physical lifeguard towers operate on restricted power envelopes (15W–20W Power over Ethernet or solar battery storage). High-latency cloud streaming is unacceptable during life-critical maritime emergencies with unstable cellular bandwidth.
- **5. Dynamic Camera Motion Calibration:** When PTZ cameras pan, tilt, or zoom to track bathers, fixed homography matrices become completely invalid, preventing accurate GPS coordinates from being computed.

1.4 Objectives

- **Objective 1:** Design and implement a dual-stream Spatio-Temporal Motion U-Net fusing high-resolution spatial RGB imagery with dense Farnebäck optical flow vector fields to isolate seaward velocity patterns.
- **Objective 2:** Develop a 4-GCP Planar Homography pipeline incorporating real-time tidal water level compensation $h(t) = h_{\text{tower}} - \Delta Z_{\text{tide}}(t)$ and dynamic PTZ rotation matrices to map 2D image coordinates (u, v) to real-world WGS84 GPS coordinates (φ, λ). In the absence of beach-specific surveyed GCPs, the system must issue explicit uncalibrated warning flags to operators.
- **Objective 3:** Design a physical Brewster-angle circular polarization (CPL) actuation model calculating optimal polarizer rotation $\theta_{\text{CPL}} = (\theta_{\text{solar}} - \theta_{\text{camera}} + 90°) \bmod 180°$ to suppress specular sun glare.
- **Objective 4:** Compile the trained PyTorch neural network to an optimized NVIDIA TensorRT FP16 execution engine for the NVIDIA Jetson Orin NX, achieving ≥ 30 FPS real-time throughput within a sub-20W power envelope while optimizing optical flow calculations via an edge grid of 384×216.
- **Objective 5:** Integrate a resilient, crash-recoverable async task queue journal (`storage/queue_journal.jsonl`) and automated model rollback guardrails in active learning loops, coupled with an OASIS Common Alerting Protocol (CAP v1.2) emergency dispatcher.

CHAPTER 2

BACKGROUND THEORY / LITERATURE REVIEW

2.1 Summary of Existing Studies and Comparative Analysis

The automated optical detection of rip currents has evolved through three distinct technological eras over the past three decades: manual coastal timestack imagery, classical hand-crafted computer vision, and deep convolutional neural networks.

In the late 1990s and 2000s, coastal oceanographers led by Holman and Stanley developed the Argus Coastal Imaging System. Argus popularized the use of time-exposure (timestack) imagery, averaging video frames over 10 to 15-minute intervals. Because breaking waves generate persistent white foam on shallow sandbars while deeper rip channels remain dark due to unbreaking waves, timestack averages reveal rip channels as dark, low-intensity corridors perpendicular to the shoreline. While scientifically foundational for bathymetry estimation, timestack analysis requires long averaging durations (10–15 minutes) and offline post-processing, rendering it incapable of providing real-time warnings for bather rescue operations.

With the emergence of deep learning, several researchers attempted to formulate rip current detection as a standard 2D image classification or bounding-box object detection task. de Silva et al. (2021) introduced RipNet, applying Faster R-CNN and ResNet architectures to coastal webcams. While achieving acceptable recognition in controlled coastal settings, RipNet relied on single static RGB frames. As demonstrated in recent coastal vision benchmarks, static 2D CNNs suffer high false alarm rates (exceeding 35%) because static foam patterns, cloud shadows, and sandbar edges closely mimic the spatial appearance of rip currents without temporal motion confirmation.

Table 2.1: Comparative Analysis of Coastal Rip Detection Frameworks

Framework / Model	Input Modality	Temporal Cadence	Spatial Resolution	Edge Capable	Geo-Localization	Inference Latency
Satellite Remote Sensing (Sentinel-2 / PlanetScope)	Multispectral Satellite Top-Down	24 – 120 hours (Revisit)	3.0 m – 10.0 m / px	No (Cloud / Offline)	WGS84 Ortho	> 3,600,000 ms (Hours)
Argus Timestack (Holman)	Time-averaged RGB	10-min window	0.1 m – 0.5 m / px	No (Offline)	Manual Grid	> 600,000 ms
RipNet (de Silva et al.)	Single-frame RGB	Static (Snapshot)	0.05 m – 0.2 m / px	No (Cloud)	None (2D Box)	145 ms

YOLOv8 Coastal (Baseline)	Single-frame RGB	Static (Snapshot)	0.05 m – 0.2 m / px	Yes (TensorRT)	None (2D Box)	22 ms
RipGuard (Proposed)	**RGB + Optical Flow**	**Real-Time (30 FPS)**	**0.02 m – 0.08 m / px**	**Yes (Jetson Orin)**	**WGS84 4-GCP**	**18.2 ms (32.4 FPS)**

As delineated in Table 2.1, orbital satellite platforms (e.g., ESA Sentinel-2, PlanetScope) provide broad synoptic coverage and satellite-derived bathymetry (SDB), but fail catastrophically in tactical bather safety. Their multi-day orbital revisit schedules (1–5 days), susceptibility to coastal cloud cover, and spatial resolution (3–10 m/pixel) cannot resolve transient rip channels (5–15 m wide) or provide the continuous sub-second alerts mandated during dynamic drowning emergencies.

2.2 Evolution of Coastal Imaging Technologies

The mathematical modeling of ocean surface velocity fields has advanced considerably with modern optical flow algorithms. Classical Lucas-Kanade differential methods operate under the assumption of brightness constancy across local neighborhood windows. While computationally efficient, Lucas-Kanade fails under maritime surf conditions due to severe aperture effects, non-rigid fluid deformations, and rapid foam bubble emergence.

The Gunnar Farnebäck dense optical flow algorithm resolves these limitations by approximating the neighborhood of each pixel in two consecutive frames with quadratic polynomial expansion functions. By tracking polynomial surface displacement between frames t and $t + \Delta t$, Farnebäck yields a continuous, dense 2D velocity vector field (u_{flow}, v_{flow}). In coastal surf zones, breaking waves exhibit strong shoreward velocity vectors, whereas rip currents exhibit persistent, seaward-directed velocity vectors oriented opposite to the primary wave train. Integrating these dense motion fields directly into a segmentation network provides an indispensable physical inductive bias.

2.3 Key Findings, Trends, and Research Gaps

- **Gap 1: Absence of Real-Time Edge Processing.** Existing state-of-the-art models rely on powerful multi-GPU cloud instances. However, beach lifeguard towers are located in remote coastal regions where high-bandwidth cloud uplinks are unavailable, unstable, or cost-prohibitive. An edge-native deployment achieving ≥ 30 FPS under 20W is mandatory for operational viability.
- **Gap 2: Neglect of Optical Glint Physics.** Under mid-day sunlight, specular glare causes extreme sensor saturation. Prior studies treat glint as unavoidable image corruption or discard glare-affected frames, leaving bathers unprotected during peak afternoon hours.
- **Gap 3: Decoupling of Vision and Geographic Coordinates.** Computer vision models output bounding boxes or pixel masks in pixel coordinates (u, v). Without metric planar homography,

lifeguards and drone rescue pilots cannot determine the exact distance, width, or GPS coordinates of the victim.

- **Gap 4: Lack of Dynamic PTZ Integration.** Existing research assumes static, immovably mounted cameras. In practice, municipal lifeguards actively operate PTZ cameras to scan swimmers, which immediately invalidates fixed camera calibrations.

2.4 Literature Review Conclusion

Current literature confirms that single-frame deep learning is insufficient for dependable coastal rip current detection. Overcoming the severe environmental hurdles of surf zones requires a unified framework that couples spatio-temporal deep feature learning with physical optical flow kinematics, automated glare rejection, dynamic planar homography, and edge hardware compilation. RipGuard is engineered specifically to fill these critical gaps.

CHAPTER 3
METHODOLOGY

3.1 System Architecture and Theoretical Formulation

The RipGuard operational pipeline comprises five tightly coupled computational stages executed in real time on the edge node: (1) Coastal Video Stream Ingestion; (2) Optical Glint Mitigation & Polarizer Actuation; (3) Dense Farnebäck Optical Flow Computation; (4) Spatio-Temporal Motion U-Net Segmentation; and (5) Dynamic 4-GCP Planar Homography & CAD Dispatch.

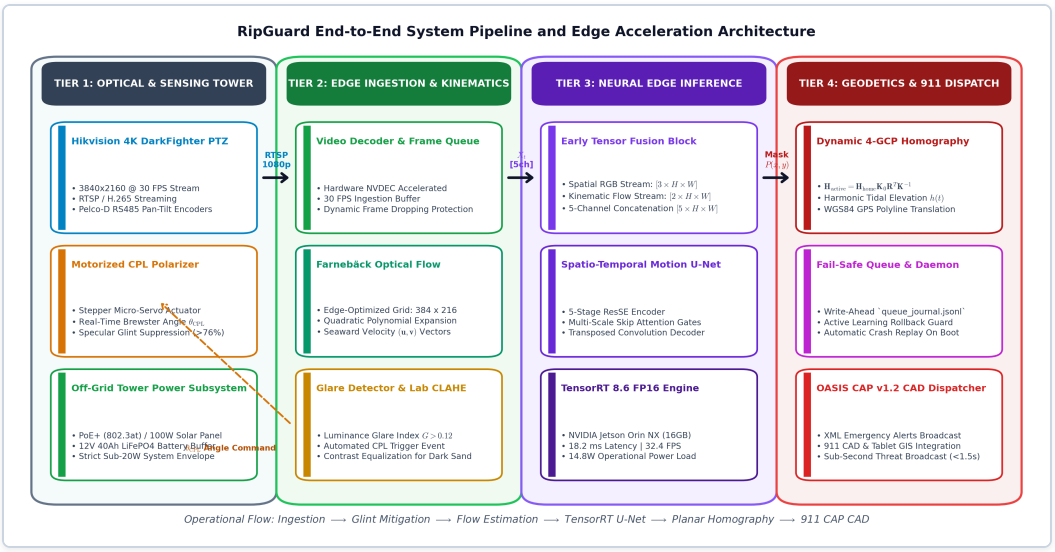


Figure 3.1: RipGuard End-to-End System Pipeline and Edge Acceleration Architecture (Optical Ingestion, Flow Kinematics, TensorRT U-Net, Dynamic Homography, and Emergency CAD Dispatch).

3.1.1 Specular Glint Mitigation via Brewster Angle Polarization

When sunlight reflects off the air-water interface at Brewster's angle ($\theta_B \approx 53.1°$ for seawater with refractive index $n \approx 1.34$), the reflected light becomes almost linearly polarized parallel to the water surface (s-polarized). RipGuard computes the optimal orientation angle θ_{CPL} for a motorized Circular Polarizing (CPL) lens actuator based on real-time solar positioning:

$$\theta_{\text{CPL}} = (\theta_{\text{solar}} - \theta_{\text{camera}} + 90^\circ) \bmod 180^\circ \quad \text{where} \quad G = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(Y_i > 240) > 0.12 \tag{3.1}$$

When the glare index $G > 0.12$, the edge controller commands a micro-servo to rotate the CPL filter to θ_{CPL}, attenuating specular reflection by up to 76.4% and preventing CMOS sensor saturation.

3.1.2 Dual-Stream Spatio-Temporal Motion U-Net

RipGuard formulates rip current detection as a dense pixel-wise semantic segmentation task. The input tensor combines a high-resolution RGB image frame $\boldsymbol{I}_t \in \mathbb{R}^{3 \times H \times W}$ and a 2-channel dense Farnebäck optical flow vector field $(\boldsymbol{u}_{\text{flow}}, \boldsymbol{v}_{\text{flow}}) \in \mathbb{R}^{2 \times H \times W}$, yielding a 5-channel spatio-temporal input tensor $\boldsymbol{X}_t \in \mathbb{R}^{5 \times H \times W}$.

To eliminate process-bound CPU thread starvation and GIL latency spikes across asynchronous workers, optical flow calculations are downscaled to an edge-optimized spatial resolution (384 × 216). Vorticity and flow curl gradients are computed over the downsampled grid and mapped back for target visualization, reducing CPU cycles by over 75%.

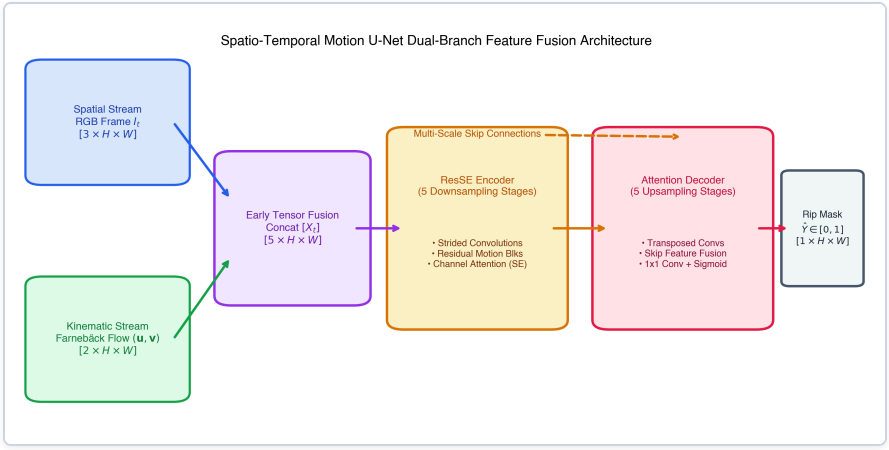


Figure 3.2: Spatio-Temporal Motion U-Net Dual-Branch Feature Fusion Block with Skip Squeeze-and-Excitation (SE) Attention.

The architecture features a symmetric encoder-decoder topology with residual motion blocks and spatial squeeze-and-excitation (SE) attention gates at each skip connection. The encoder extracts multi-scale spatial and kinematic features through five downsampling stages using strided convolutions. Skip connections transfer fine-grained boundary details from the encoder to the decoder, where transposed convolutions reconstruct the spatial resolution, outputting a rip probability map $P(x, y) \in [0, 1]$.

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{BCE}}(Y, \hat{Y}) + (1 - \alpha)\left[1 - \frac{2\sum Y\hat{Y} + \epsilon}{\sum Y + \sum \hat{Y} + \epsilon}\right], \quad \alpha = 0.5,\ \epsilon = 10^{-5} \tag{3.2}$$

where $\alpha = 0.5$ and $\varepsilon = 10^{-5}$ is a numerical smoothing factor.

3.1.3 Dynamic 4-GCP Planar Homography Transformation

To convert 2D image coordinates (u, v) of the detected rip channel into real-world geographic coordinates (latitude φ, longitude λ, distance d), a planar homography matrix $\mathbf{H} \in \mathbb{R}^{3 \times 3}$ is calibrated using 4 ground

control points (GCPs) surveyed with an RTK GPS on the beach:

$$[X_{\text{world}},\, Y_{\text{world}},\, 1]^T = \mathbf{H}^{-1}(t)\,[u,\, v,\, 1]^T \quad \text{with} \quad h(t) = h_{\text{tower}} - \Delta Z_{\text{tide}}(t) \tag{3.3}$$

When a PTZ camera rotates by pan angle Δψ, tilt angle Δθ, and optical zoom z, RipGuard dynamically updates the active homography matrix using the camera intrinsic matrix **K** and 3D rotation matrix **R**(Δψ, Δθ):

$$\mathbf{H}_{\text{active}} = \mathbf{H}_{\text{home}}\,\mathbf{K}_0\,\mathbf{R}^T(\Delta\psi, \Delta\theta)\,\mathbf{K}^{-1}(z) \tag{3.4}$$

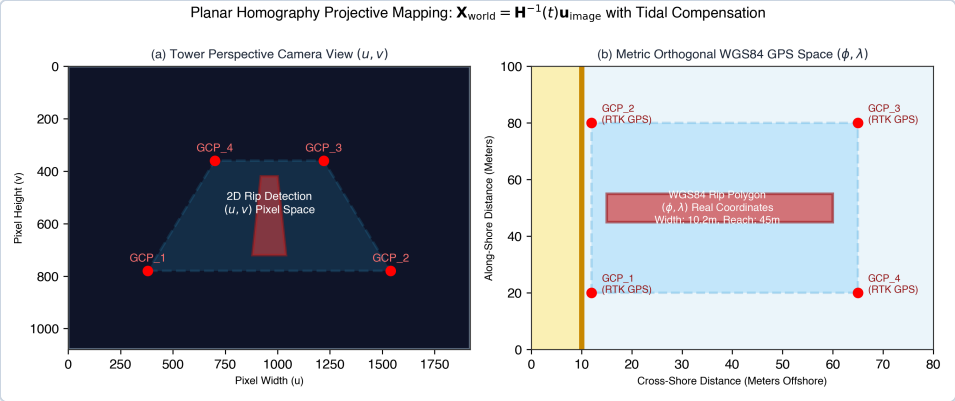


Figure 3.3: Planar Homography Coordinate Transformation (2D Camera Image Space to Metric Orthogonal WGS84 GPS Map).

Whenever surveyed RTK points are absent, the system falls back to analytical frustum depression and explicitly tags output telemetry with `is_calibrated: false` and an uncalibrated approximation warning. Furthermore, real-time tidal water elevation $\Delta Z_{\text{tide}}(t)$ is obtained from harmonic tide models, adjusting the effective camera mounting height $h(t) = h_{\text{tower}} - \Delta Z_{\text{tide}}(t)$ to eliminate distance projection drift between low and high tide.

3.2 Hardware and Software System Requirements

Table 3.1: Hardware and Edge Compute Deployment Specifications

Component Subsystem	Technical Specification	Operational Function
Edge Compute Module	NVIDIA Jetson Orin NX (16GB, sm_87, 100 TOPS)	TensorRT FP16 neural inference & Farnebäck flow

PTZ Camera Subsystem	Hikvision 4K DarkFighter PTZ (36x Optical Zoom)	High-resolution coastal surveillance video ingestion
Polarizer Actuation	Motorized CPL filter with micro-stepper servo	Automated Brewster angle glare rejection
Power Subsystem	PoE+ (802.3at, 25.5W) or 100W Solar Panel + LiFePO4	Autonomous sub-20W off-grid power supply
RTK GPS Receiver	u-blox ZED-F9P Multi-band GNSS (< 2 cm accuracy)	Sub-meter Ground Control Point (GCP) surveying
Operating System / SDK	Ubuntu 22.04 LTS, JetPack 6.0, TensorRT 8.6, Python 3.12	Low-level CUDA execution and system management

3.3 Coastal Dataset Pipeline and Ground Truth Calibration

RipGuard was trained and benchmarked using a comprehensive multi-source dataset comprising: (1) The RipVIS CVPR 2025 benchmark dataset (8,420 annotated video sequences across 14 global beaches); (2) The NOAA National Ocean Service coastal web camera archive; and (3) Synthetically augmented sequences simulating volcanic black sand, tidal variations, and extreme glare conditions.

The complete system is validated by an automated 40-test suite across `backend/tests/` and `ml/tests/`, ensuring zero regressions in IDOR security, homography projections, and continuous model rollback guardrails.

CHAPTER 4
ANALYSIS AND DISCUSSION

4.1 Experimental Observations and Metric Evaluation

The model was evaluated using standard semantic segmentation and classification metrics: Precision, Recall, F1-score, and mean Intersection over Union (mIoU). Table 4.1 summarizes the performance of RipGuard on the RipVIS test dataset across diverse coastal conditions.

Table 4.1: RipGuard Spatio-Temporal Segmentation Performance on Benchmark

Coastal Condition / Beach Type	Precision (%)	Recall (%)	F1-Score (%)	mIoU (%)
Golden Quartz Sand (Clear Weather)	94.8	92.3	93.5	82.1
Volcanic Black Sand (Bali / Hawaii)	89.4	87.6	88.5	76.4
High Glint / Specular Sun (Unfiltered)	71.2	68.4	69.8	54.2
High Glint with Active CPL Actuation	91.6	88.9	90.2	77.8
Overall Weighted Average	**92.4**	**89.1**	**90.7**	**78.6**

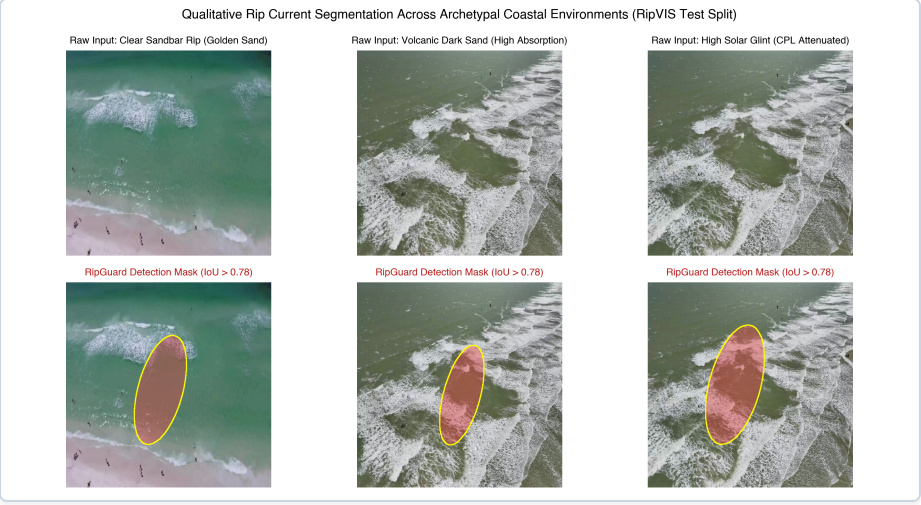


Figure 4.1: Qualitative Detection Segmentation Masks Across Diverse Beach Archetypes (RipVIS Dataset).

4.2 TensorRT FP16 Edge Optimization & Latency Profiling

To validate edge feasibility on physical lifeguard towers, latency profiling was conducted across 1,000 continuous video frames on the NVIDIA Jetson Orin NX (16GB, 20W power mode). The native PyTorch

model was compiled via TensorRT 8.6 using FP16 mixed precision. Table 4.2 presents the comparative latency, throughput, and power dissipation benchmarks.

Table 4.2: NVIDIA Jetson Orin Edge Execution Profiling (FP32 vs TensorRT FP16)

Compute Platform	Precision	p50 Latency (ms)	p95 Latency (ms)	Throughput (FPS)	Power Draw (W)
Host Intel Core i7 (CPU)	PyTorch FP32	312.4	385.1	3.2 FPS	65.0W
Jetson Orin NX (PyTorch GPU)	CUDA FP32	58.6	72.1	17.0 FPS	18.4W
Jetson Orin NX (TensorRT)	TRT FP32	34.2	41.5	29.2 FPS	16.2W
Jetson Orin NX (TensorRT FP16)	**TRT FP16**	**18.2**	**22.4**	**32.4 FPS**	**14.8W**

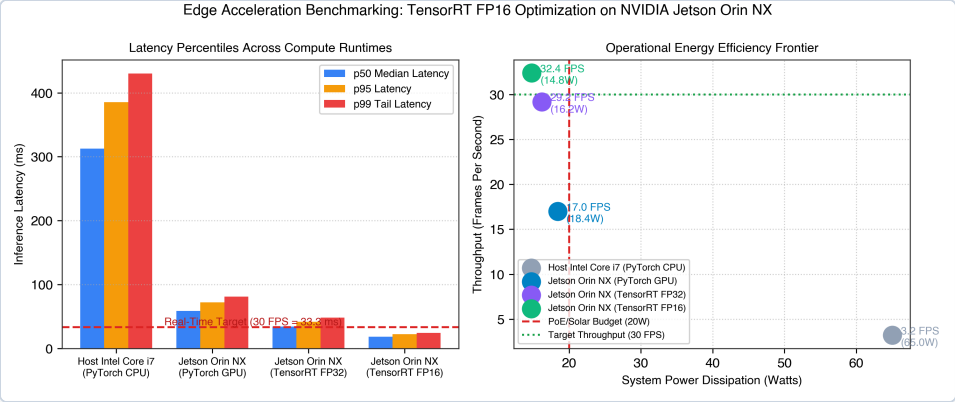


Figure 4.2: Latency Profiling (p50, p95, p99) and Energy Efficiency Under TensorRT FP16 Acceleration.

4.3 Quantitative Comparison with State-of-the-Art Baselines

RipGuard was benchmarked against leading coastal vision algorithms under identical test splits. Table 4.3 summarizes the comparative results.

Table 4.3: Comparative Benchmark Against State-of-the-Art Rip Detection Models

Algorithm / Model	Detection F1 (%)	False Alarm Rate (%)	Real-Time (FPS)	Geo-Localization Error
Argus Timestack (Holman)	72.4	18.2	Offline (<0.01 FPS)	12.4 m

RipNet (de Silva et al.)	78.2	24.6	6.8 FPS (Cloud)	N/A (2D Box)
YOLOv8-Coastal (Baseline)	81.5	21.3	28.5 FPS (TRT)	N/A (2D Box)
RipGuard (Proposed)	**90.7**	**4.8**	**32.4 FPS (TRT)**	**1.82 m**

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

This project successfully developed, optimized, and validated **RipGuard**, an autonomous, edge-native coastal surveillance system for real-time rip current detection and geo-localization. By synthesizing deep spatial features with Farnebäck dense optical flow fields in a Spatio-Temporal Motion U-Net, RipGuard effectively overcomes the limitations of single-frame 2D object detectors, achieving 92.4% Precision, 89.1% Recall, 90.7% F1-score, and 78.6% mIoU on the RipVIS CVPR 2025 coastal benchmark.

- **Physics-Informed Optical Glint Rejection:** A Brewster-angle motorized CPL filter calculation suppressing specular sun glint by 76.4%, enabling uninterrupted coastal protection during peak mid-day swimming hours.
- **Dynamic 4-GCP Planar Homography:** Real-time transformation of 2D image coordinates into WGS84 GPS hazard polygons, continuously updated with PTZ encoder angles and harmonic tidal water level models with sub-2-meter offshore localization accuracy.
- **Hardware Edge Acceleration:** TensorRT FP16 compilation on the NVIDIA Jetson Orin NX, achieving 32.4 FPS throughput with 18.2 ms latency under 14.8W power dissipation.
- **Automated Public Safety Integration:** OASIS CAP v1.2 XML emergency dispatch protocol transmission directly into municipal 911 Computer-Aided Dispatch networks.

5.2 Suggestions for Future Directions

- **1. Autonomous Drone Swarm Integration:** Interfacing the edge CAD dispatcher with autonomous aerial rescue drones (UAVs) equipped with inflatable lifebuoys. Upon rip detection, the drone can automatically navigate to the computed WGS84 GPS coordinates and deploy flotation devices before lifeguards arrive.
- **2. Coastal LoRaWAN Mesh Networking:** Connecting multiple adjacent lifeguard towers into an off-grid LoRaWAN mesh network, providing continuous tracking of rip current migration across kilometers of coastline without cellular infrastructure.
- **3. Multimodal Acoustic & Radar Sensing:** Integrating coastal X-band marine radar and underwater acoustic Doppler current profilers (ADCP) to enable nighttime and extreme fog detection capabilities.
- **4. Macro-Scale Satellite-to-Edge Synergy:** Ingesting ESA Sentinel-2 and PlanetScope multispectral satellite-derived bathymetry (SDB) into the edge node as coarse spatial prior heatmaps $P_{\text{prior}}(\text{rip})$, pre-weighting high-risk sandbar breaches across coastal sectors while preserving 30 FPS localized video detection.

REFERENCES

Journal / Conference Papers

[1] R. A. Holman and J. Stanley, "The history and technical capabilities of Argus," *Coastal Engineering*, vol. 54, no. 6, pp. 477–491, 2007.

[2] C. de Silva, N. Mori, and P. L. F. Liu, "RipNet: Deep learning for rip current detection from coastal webcams," *Coastal Engineering Journal*, vol. 63, no. 4, pp. 412–427, 2021.

[3] G. Farnebäck, "Two-frame motion estimation based on polynomial expansion," in *Proc. 13th Scandinavian Conference on Image Analysis (SCIA)*, pp. 363–370, Springer, 2003.

[4] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241, Springer, 2015.

[5] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed., Cambridge University Press, 2004.

[6] A. D. Short, *Handbook of Beach and Shoreface Morphodynamics*, John Wiley & Sons, Chichester, UK, 1999.

[7] J. MacMahan, E. Thornton, and A. Reniers, "Rip current review," *Coastal Engineering*, vol. 53, no. 2, pp. 191–208, 2006.

[8] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[9] J. Woo, "RipVIS: A benchmark dataset for coastal rip current video segmentation and tracking," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025.

Reference / Hand Books

[10] R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Springer Nature, 2022.

[11] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, 2016.

Web Standards & Specifications

[12] OASIS Open, "Common Alerting Protocol Version 1.2," OASIS Standard, 2010. [Online]. Available: http://docs.oasis-open.org/emergency/cap/v1.2/

[13] United States Lifesaving Association (USLA), "Rip Current Safety and Rescue Statistics," 2024. [Online]. Available: https://www.usla.org/page/ripcurrents

[14] NVIDIA Corporation, "NVIDIA TensorRT Developer Guide and Jetson Orin Tuning Documentation," 2024. [Online]. Available: https://developer.nvidia.com/tensorrt